Session 3 - Effects & Processing

1. Audio Effects

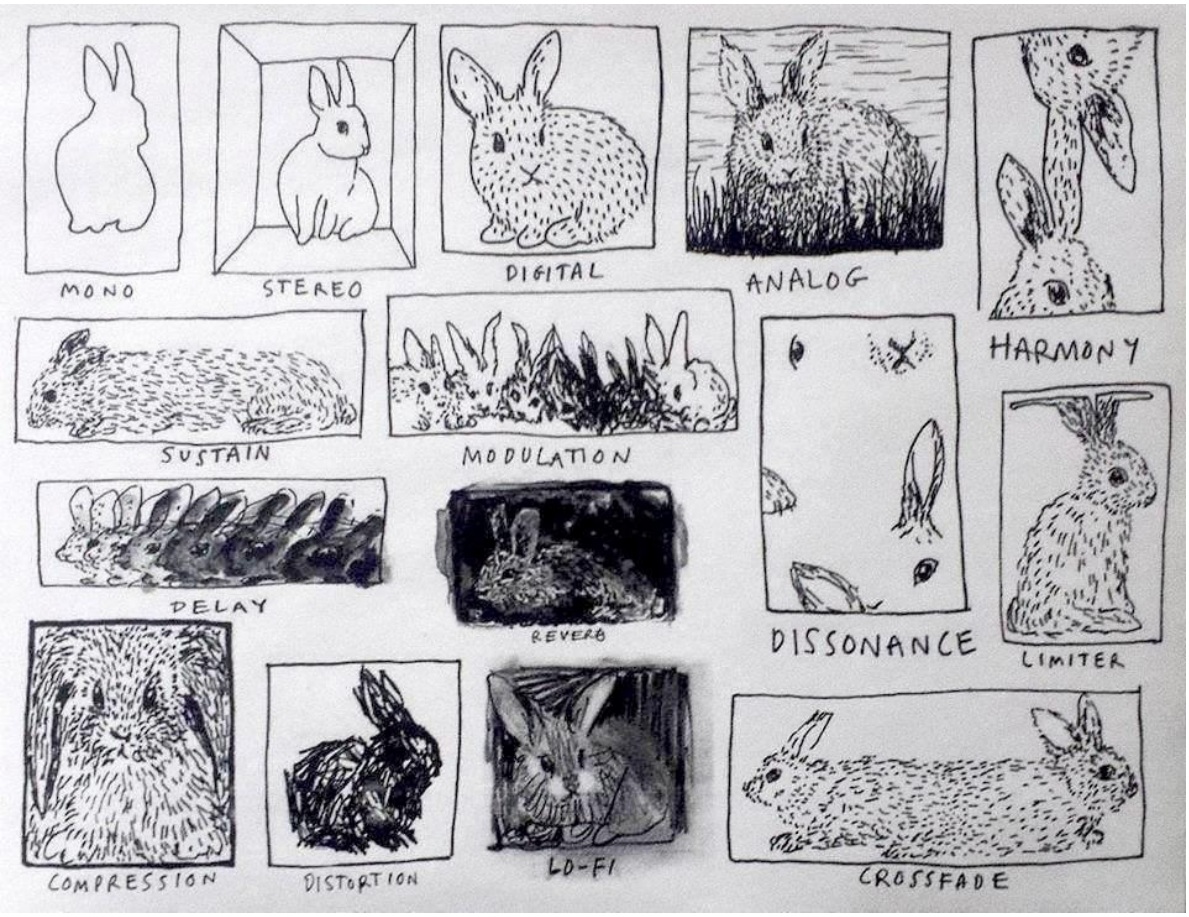

fig. 1 A visual representation of common audio effects and qualities

i. Reverb

Reverb is one of the most ubiquitous and iconic effects in music, and Ableton's Reverb device is an excellent choice for adding quick reverb to your tracks.

Reverb is commonly used to:

- Give a sense of physical space to an instrument or group of instruments

- Create large ambient washes of blurred sound
- Provide quick chorused slapback effects on percussion

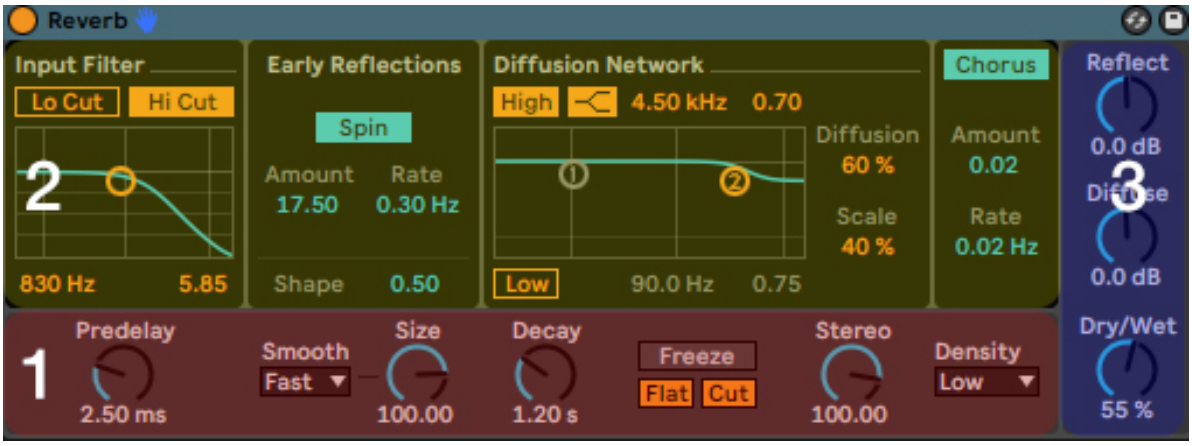

fig. 2 The sections of Reverb's interface

Reverb can broadly be broken up into the following sections:

1. Spatial controls
2. Filter controls
3. Shaping controls

When working with Reverb, the spatial controls are most often what you'll be using. **Predelay** can be used to give a small amount of time from the start of a sound before the reverb kicks in — this is used when putting reverb on something percussive to allow the initial transient to still punch through. **Size** controls how large the artificial space of the reverb is, while **decay** sets how long the reverb rings out for.

The **filter controls** are more advanced and nuanced. The most important feature here is the **input filter**, which lets you choose what part of your sound is receiving the reverb. It is common practice to use a highpass ("Lo Cut") set at around 200Hz so no low signal gets through, while also rolling off highs with a lowpass around 10-15kHz ("Hi Cut") to keep harsh bright tones from being reverberated. In both cases, this can clean up the reverb signal and keep it from becoming muddy.

In the **shaping controls**, the **Dry/Wet** knob is the most essential parameter along with the spatial controls. Dry/Wet is a parameter we will see in nearly all effects going forward, and allows us to choose how much of the unaffected vs. affected signal is

being passed through. At 0%, we are getting 100% of the original signal coming through and no processing, while 50% will allow you to hear both the processed and unprocessed signal equally, and 100% will be fully processed. This will regularly be the most important parameter for shaping your sound overall.

While you can add reverb directly onto your tracks, it is best practice to instead put a single Reverb device on a **return track**.

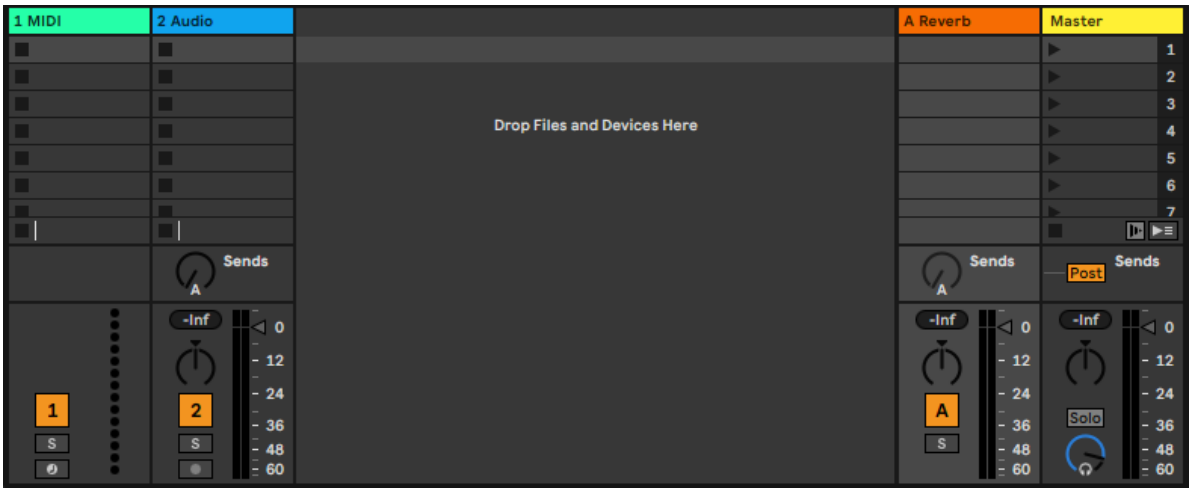

fig. 3 Return track “A Reverb” containing a Reverb device. While normal tracks are numbered sequentially (1, 2, 3, etc.), return tracks are lettered (A, B, C, etc.) sequentially

Return tracks, called **buses** in other software, are a special kind of track that allow you to send the sound from audio and MIDI tracks into them and process all these tracks together. You can think of it like a bus that carries any tracks you put into it to your master channel. Return tracks are best used when there’s an effect chain you’ll be using repeatedly throughout your track to cut down on the computational power of having an individual signal chain for each audio or MIDI track. This is very commonly done with reverb, delay, and as we’ll later see today, parallel compression.

Make a new return track (CTRL/CMD + ALT + T), and notice you’ll have a new track that appears next to your Master channel. Drop a Reverb here and set the “Dry/Wet” to 100%. We want 100% because we’ll be using **sends** to control how much of each track’s signal gets sent to the return track. In this way, sends act like a dry/wet for each individual track we send to our return track.

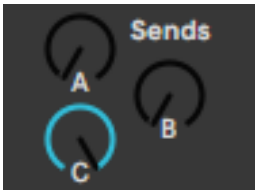

fig. 4 Send C turned all the way up so the entire signal is being effected

ii. Mixed Signals with Hybrid Reverb

One of the newer devices in Ableton, hybrid reverb is a fascinating reverb device with two distinct forms of reverb. The first, **algorithmic**, functions similar to our standard reverb, using different algorithms to delay and shape your input. Controls for algorithmic reverb are located on the bottom right of hybrid reverb.

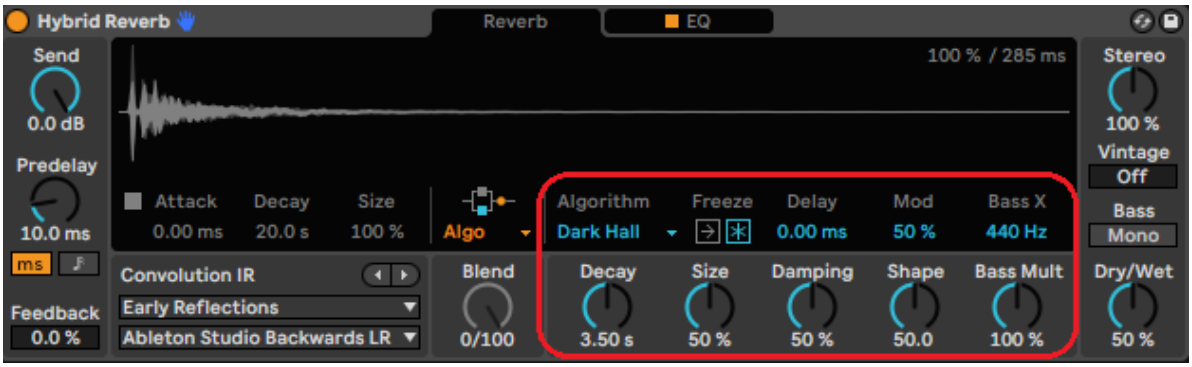

fig. 5 The controls for shaping algorithmic reverb. The dropdown under “Dark Hall” selects the algorithm, and the controls below are used to shape it.

The second form is **convolution.** Convolution reverb is a special type of reverb that uses cross-synthesis to combine its input signal with another signal, called an impulse response or IR.

Ordinarily this IR might be something like the sound of a room or hall, so that our input signal emphasizes only the frequencies it shares with this room impulse. This will make our input sound like it is a part of this space. This can give a natural sense of space to a sound and glue different sounds together into a single ‘space’.

However, the magic of convolution reverb comes when we use unorthodox impulse responses. Instead of the sound of a room, what if we use foley, noise, vocals, glitches, synthesis? Because convolution works by emphasizing the frequencies an input shares

with this convolution, we can make one sound take on sonic the characteristics of any other sound.

Hybrid reverb comes preloaded with a good variety of IRs already, which can be found under the “Convolution IR” dropdown. The waveform shows our selected IR.

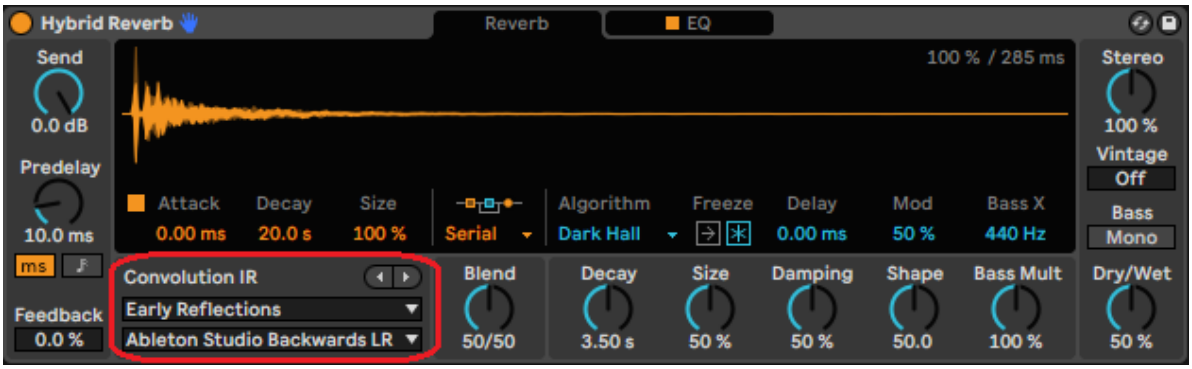

fig. 6 Hybrid reverb. Convolution IR contains impulse responses.

These IRs can be blended with our input signal using the routing configuration and blend knob.

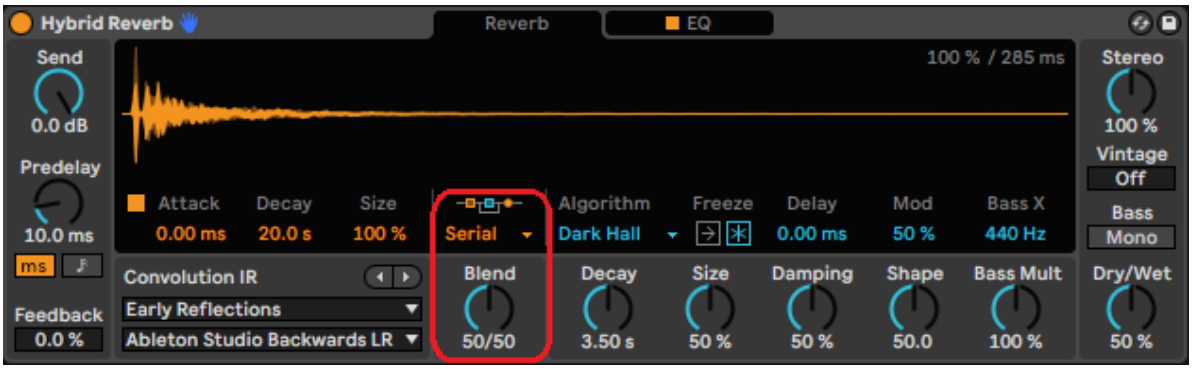

fig. 7 Routing configuration located in the “serial” dropdown. Blend 100/0 is 100% convolution.

Serial and parallel routing use both convolution and algorithmic reverb. Algorithmic and convolution can be selected to use their respective mode alone. The blend knob will select the amount of convolution / algorithmic reverb being used when parallel or serial modes are selected.

The most powerful feature of hybrid reverb is the ability to load in user IRs. This can be done by clicking, dragging, and dropping an audio file over the waveform view in hybrid reverb. These files can come from within or outside Ableton.

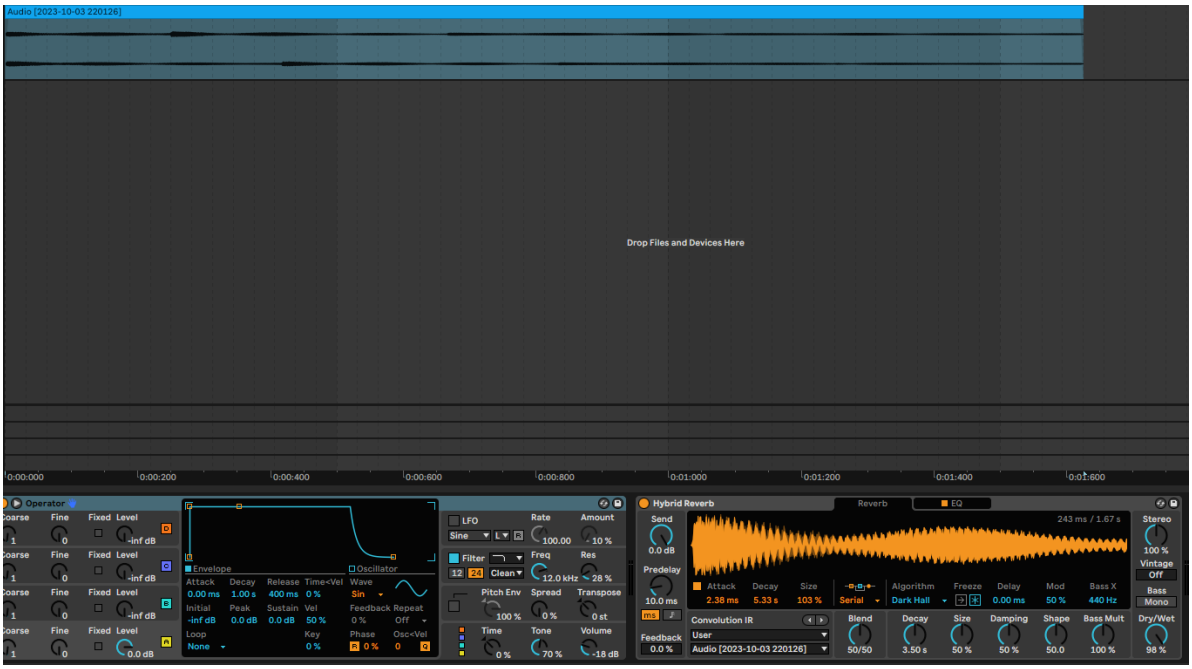

fig. 8 A note from an Impulse instrument was played in a MIDI track, recorded to an audio track, then dragged into hybrid reverb to serve as an IR for this Operator.

There's no limit to the sound design possibilities that can come from this technique. Ever wanted your synth pad to be more metallic? Use a spring being pulled or metal clank foley as an IR. Ever wondered what your vocals would sound like if they were a hybrid of your bassline? Record the bassline and use it as an IR.

Think about what you can do by using one IR, resampling, then using another IR, then resampling and using another…

iii. Modulation with LFO and Shaper

The recent addition of the **Modulator effects** into Ableton completely revolutionized the way many people, including myself, work with Ableton. While there exists three: LFO, Shaper, and Envelope Follower, we will only be taking a look at LFO and Shaper today for the sake of time.

LFO is a universal LFO that can be mapped to any parameter in any Ableton device.

fig. 9 LFO's interface

Working with LFO is very easy. Simply load an LFO into the same effect chain of whatever parameter you want to control, then click the **"Map" button** in the top left. The button will start blinking, showing it's ready to map. Then click on the parameter you want to map to. You should automatically see the parameter moving.

Now just shape your LFO to taste. Use the two percentages in the top right to set a lower and upper bound, **"Rate"** to control speed, **"Depth"** to control the range of the parameter, and "**Offset"** to shape it.

Underneath the waveform you can use the leftmost dropdown to select different waveforms. To get **randomized automation of any parameter**, set the waveform to **"Random"**. By default it will function like the Sample & Hold wave we saw last week. To get smooth, more human-like randomization, bring up the **"Smooth"** percentage to 100% to have LFO glide between its random values. Then bring up **"Jitter"** to taste, which will create a random jitter around the values it goes to. I find values of 5%-9% add a reasonable amount of organic movement.

fig. 10 Random, Smooth, and Jitter working together to create random organic movement

Shaper functions similar to LFO, but gives even more control by letting you design the waveform yourself.

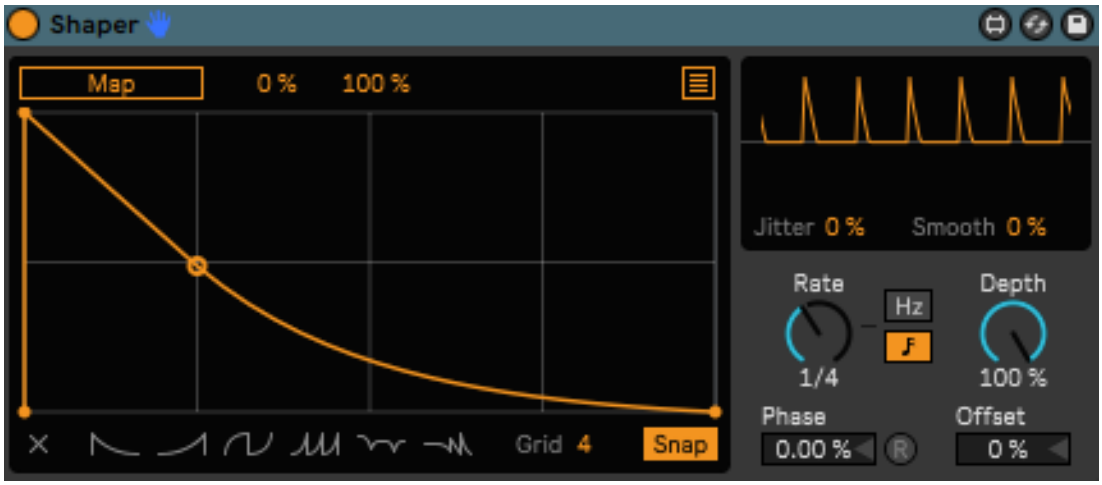

fig. 11 Shaper's interface

While the right side of Shaper acts exactly the same as LFO, we now have a **waveform view** on the left side of the device to draw in a custom waveform. Simply click within the grid to add nodes, and click and drag to reposition.

You can use the preset waveforms along the bottom by clicking them, and change how fine you can edit the waveform using the "**Grid"** parameter. Turn off the **"Snap"** button to allow you to move nodes off the grid. Hold ALT and click and drag between any two notes to change the inflection of the curve between nodes.

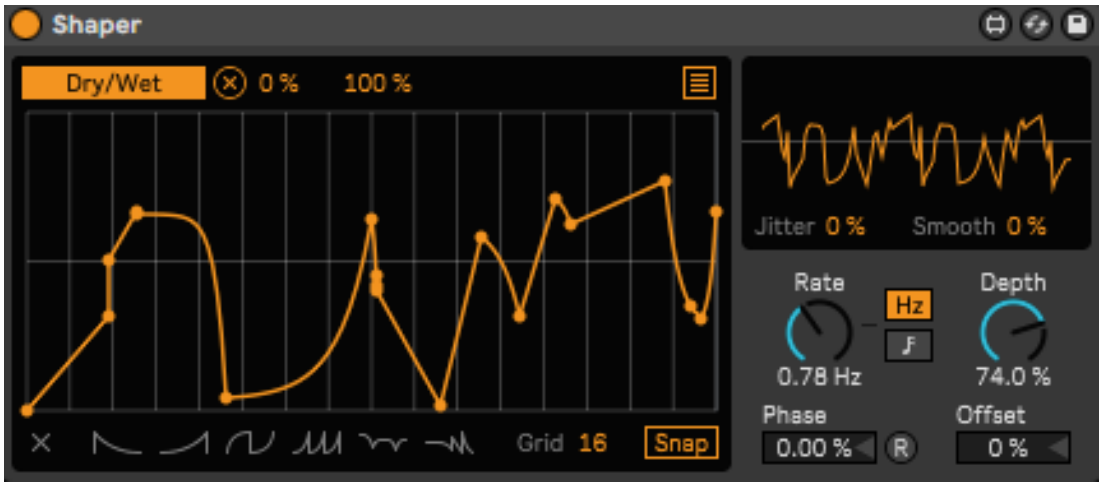

fig. 12 A chaotic, random waveform with custom inflection by drawing off the grid

iv. Endless Sounds with echo echo echo

At first glance, **echo** might seem like a simple delay effect, but it is so much more. Echo is similar to a delay pedal that will take a fragment of your sound and repeat it, but contains a wealth of parameters that let you carefully shape how your sound repeats and fades with time.

Echo can be used to:

- Create realistic spatial echo
- Add syncopation and fill out grooves on percussion
- Create thick, lush washes of sound that are less diffuse than with reverb by using large time and feedback settings
- Create strange metallic textures with very small time settings with slight modulation
- Generate chaotic SFX by turning feedback over 100%

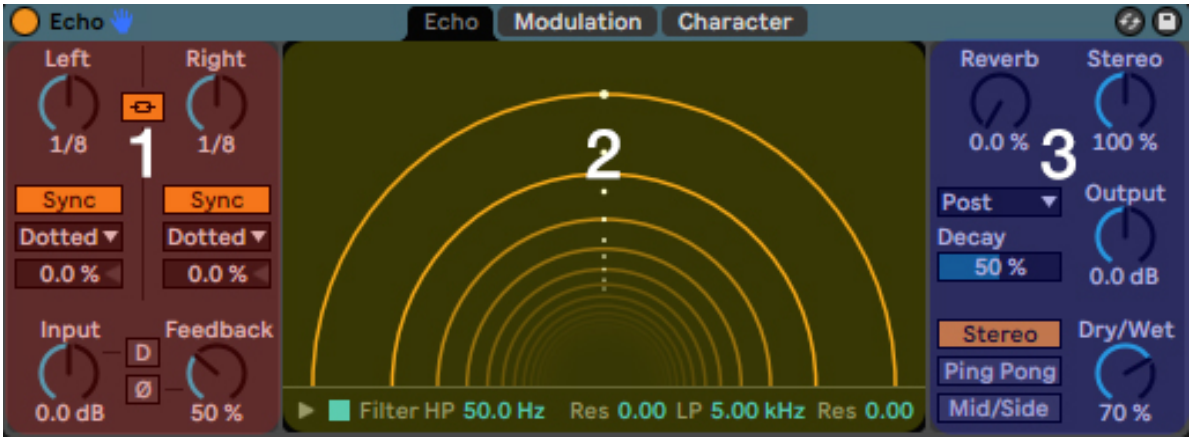

fig. 13 The main sections of Echo's interface

Echo can be broken into 3 sections:

1. Time and feedback controls
2. Graphic and filter controls
3. Output controls

In the time and feedback controls, we can use the **"Left"** and **"Right"** knobs to set the time of the echo. Click the **chain** toggle between them to link/unlink each side to the left and right can be changed separately. Click the **"Sync"** button to switch between time in notes or milliseconds. The **"Feedback"** knob controls the amount of output from Echo that is fed back into Echo's input. At 100% or more feedback, you will get in an endless loop of sound. BE CAREFUL with this parameter, as you can quickly get extremely loud feedback from going over 99% feedback.

In the graphic and filter controls, we have a graphic representation of the parameters we set in the time and feedback controls. You can set the parameter of the filter along the bottom.

The output controls have a **"Reverb"** knob to set built-in reverb, a **"Stereo"** knob to set how wide the stereo image is, a dry/wet knob, and toggle switches for three different kinds of echo. **"Stereo"** functions like a standard echo with left and right channels shared. "Ping-Pong" mixes the left and right channels separately, creating a ping-pong effect of the echo bouncing back and forth. "Mid/Side" splits the stereo image so the far sides of the image are separate from the middle.

Switching to the **“Modulation” tab** in the top middle of Echo, we can access an LFO that can be used to shape either the delay or filter by shaping the waveform you like, then setting the percentages above “Delay” and “Filter” to the amount you would like to modulate each parameter.

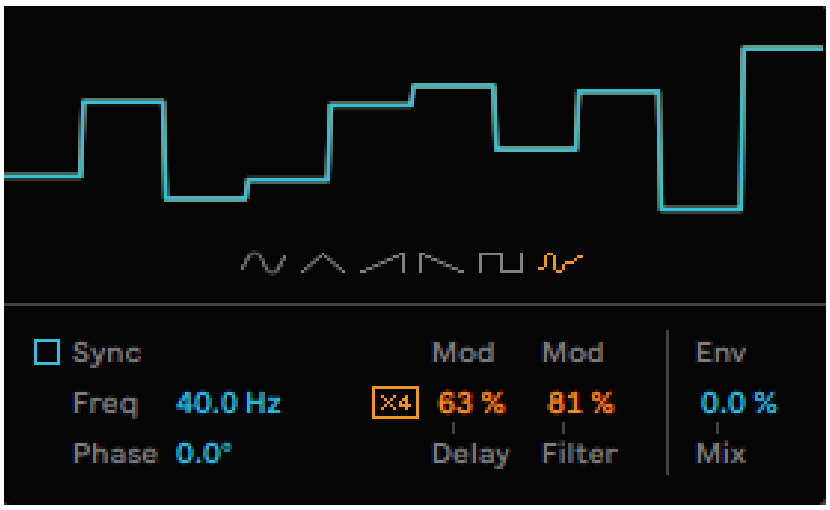

fig. 14 A random waveform modulating both delay and filter. Phase has been turn to 0.0° to effect both left/right channels at the same time

Finally, go to the **“Character” tab** to add a variety of effects. Allow only some of the signal through with **“Gate”**, duck the wet signal at a certain volume with **“Ducking”**, add noise to the wet signal with **“Noise”**, and introduce pitch modulation with **“Wobble”**.

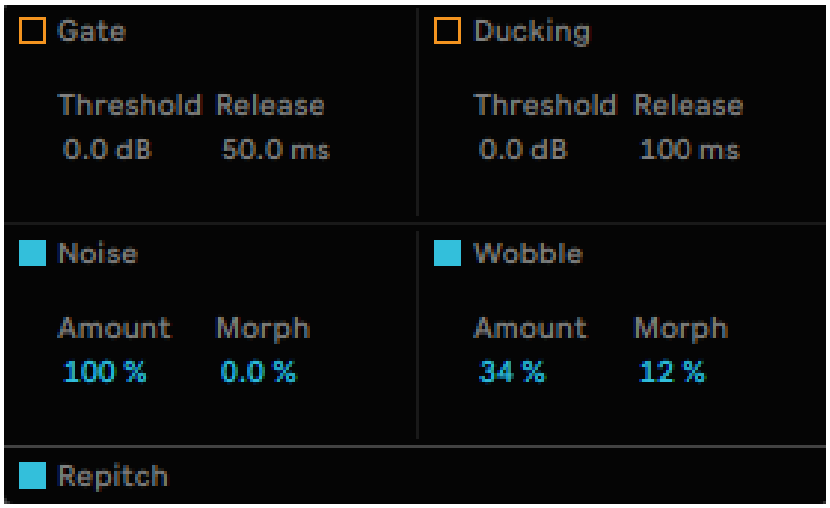

fig. 15 Character controls. Noise and Wobble are turned on

A fun technique to try is to put an Echo on an empty track, turn Noise on and set “Amount” to 100% to generate white noise. Now use the Character, Modulation, time, and feedback controls to shape and modulate the white noise into a wide variety of interesting and strange SFX without a single input source.

v. Sculpting Frequencies with EQ Eight

EQ Eight is Ableton’s default **equalizer**, and is undoubtedly the effect I use the most, having one on almost every instrument I use. Equalization in general is one of the two most essential tools in mixing, along with compression. EQ Eight works exactly the same as the filters we used to sculpt away frequencies on the instruments we looked at last week, but it is much more robust and can work on any sound.

The “Eight” in the name refers to EQ Eight having 8 total nodes (or filters) we can use to sculpt our frequency. Each filter has the same controls:

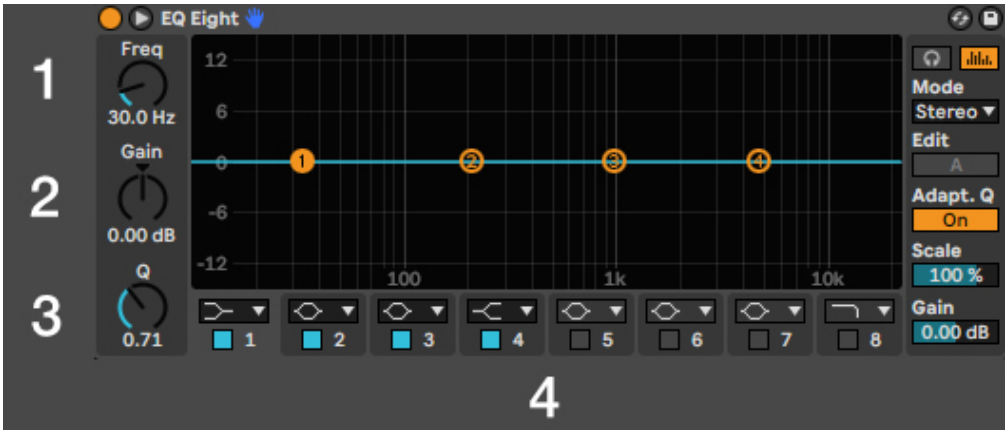

fig. 16 The primary controls of EQ Eight

1. **Freq** — Set the frequency to apply the filter around
2. **Gain** — How much to cut or boost at the selected frequency
3. **Q** — Set the bandwidth of the filter, or how much is cut/boosted around the frequency you set
4. **Filter Mode** — The shape of the filter

You can click on the corresponding node in the main **frequency display** to access the controls for that individual filter. You can also adjust these parameters by simply clicking and dragging in the display rather than using knobs.

One little known trick with EQ Eight is to right click on the top bar of the device and select **"Oversampling"**. This mode, while slightly more CPU intensive, will provide a better overall sound.

Learning to use an equalizer such as EQ Eight masterfully can take many years to train your ears and learn common conventions and is thus beyond the scope of this class. For now, I'll provide some tips on how I like to use EQ Eight and how to get comfortable with equalizers in general.

1. If you have a 'problem' frequency you'd like to remove but aren't sure where it's located, set any filter to a **notch filter** (the 5th option from the top shaped like a U in the dropdown menu). This completely cuts out frequencies around the filter. Now scan through your frequency band with the "Freq" knob until you hear that frequency disappear. Now bring up the "Q" knob to tighten the range being cut and adjust Freq to ensure you're still cutting the sound. Once you have the Q as tight as possible to completely remove the problem sound, the number on the Freq knob is where the frequency is located. If you want a less extreme cut, you can switch the filter to a bell filter (the 4th option from the top) and bring the Gain knob down to taste.
2. A general rule of thumb is to **almost always cut, don't boost**. There are some exceptions to this in more advanced sound design, but generally if you would like to hear more of the lows/mids/highs of an instrument, it's better to start by cutting around that frequency in other instruments. For example, if you want your bass to be louder around 400Hz, try gently cutting around 400Hz on other low-end instruments like your kick, or check if instruments such as your pads or leads have some unnecessary spectral content around 400Hz. If that still doesn't work, try gently cutting in other parts of your sound (such as the mids or highs in our bass example) and bringing the volume up. Only if that still isn't satisfactory should you consider boosting the Gain.
3. It's generally a good idea to **set a lowpass or highpass filter on instruments to cut away unnecessary frequencies** to keep a clean mix, such as putting a gentle lowpass above 1000-1200Hz for a bass, or a highpass around 800Hz for high hats. However, it is very easy to go overboard with this and leave your mix sounding

tinny, hollow, or lacking energy, so always think about whether your ears are telling you to use EQ, or if you're using it because you think you should.

4. Most importantly and in conjunction with the last tip, **mix with your ears, not your eyes**. Just because you might see something on the frequency spectrum that looks like it should be cut, doesn't mean it necessarily should. It's a good idea to close your eyes periodically and try to listen deeply to search for changes that need to be made. Remember that your audience won't be seeing what the spectrum looks like. "Ear fatigue" where it becomes difficult to discern frequencies well after awhile is also quite real, particularly at higher volumes, and taking periodic breaks while mixing and equalizing is necessary to keep your brain and ears fresh. If it sounds good on at least a couple different sound systems (headphones, earbuds, speakers, a car, etc.), chances are the mix is fine as is.

vi. Compressor

Ducking with Sidechain Compression

Sidechain compression is a special kind of compression that lets you duck one track out of the way whenever another track is at a certain volume.

This is one of the most important mixing techniques for most styles of modern electronic music to keep all the punch of your kicks while still keeping a heavy loud bass in the mix by keeping their lower frequencies from clashing with each other.

To do this, we put a compressor on the track we want to duck out — in this case, our bass track. Click the small arrow icon on the top left of the compressor to show the sidechain settings. From the dropdown menu, select your kick track. You can use the **EQ knob** to let only a certain part of the kick's frequency control the compressor. This can be useful if you want less ducking with a clickier (and thus higher frequency) kick.

fig. 17 Compressor with sidechain compression activated

Now play both tracks and bring down the **threshold** on the compressor until you hear the bass ducking around the kick. Use the attack and release knobs to shape how quickly the ducking starts and ends respectively. A very low threshold and/or high attack/release will produce a dramatic pumping effect.

Although kick-bass sidechain compression is the most common usage, you can sidechain anything to anything.

Try making a MIDI track with a basic Operator patch, create a MIDI track with notes of the same pitch playing a rhythm, then mute this track and sidechain it to something like a pad. You'll now hear that pad ducking to the rhythm you made with Operator without hearing Operator itself, allowing you to perform **ghost sidechaining**.

Boosting Loudness and Presence with Parallel Compression (ft. Ben Frost)

One of the most used techniques I've picked up over the years is taking advantage of Ableton's **return tracks** for over the top **parallel compression.** This is a technique used extensively by Ben Frost on his album "By the Throat" to achieve an extremely heavy, compressed, and ultimately loud quality on his mixes, and here I'll be demonstrating his exact signal chain for achieving this effect. In his project files, he calls the return track that achieves this sound "the suck", as at its extremes it can produce a similar pumping or sucking sound as sidechain compression.

Parallel compression is a technique of mixing both a dry (uncompressed) version of a sound with a wet (overly compressed) version of the same sound. The compression increases the perceived loudness and presence of the sound, while the uncompressed signal helps maintain the dynamics that are squashed out by the compressed signal.

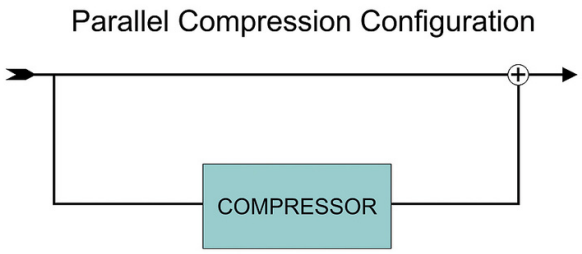

fig. 18 A diagram of parallel compression, showing the compressed and uncompressed signals being summed

To start, make a new return track and add a compressor to it. The first parameter to look at is **ratio**. Ratio is a measure of how much compression happens above the threshold we set. With typical compression, we would consider a ratio of 3:1 (for every 3dB above our threshold, we increase the output by 1dB) to be moderate. For our purposes, we're using a ridiculous ratio of **25.0:1** to allow a huge amount of compression to happen.

Attack and **release** shape how quickly the compression takes place once sound enters the compressor, and how fast it goes away, respectively. We can leave attack at around 1ms, but to get a more dramatic pumping sound, we want a short release of 3.60ms.

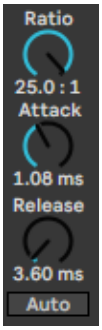

fig. 19 Ratio, attack, and release

We also want our compression to start when the sound coming in is very quiet, so we set our **threshold** to -20dB. This means that the compression (that we set with our ratio), will start when the compressor receives at least a -20dB signal.

Finally, we want to set how dramatically the compressor responds. We can do this by first changing the **knee**. When wanting subtle compression, the knee gives some wiggle room around the threshold so that it will gradually kick in as the loudness of the incoming signal approaches the threshold. Because we are going for a dramatically compressed sound, we want to set the knee to the highest setting: 0dB. This is known as a **hard knee**, and means that compression will only start exactly at threshold.

Similarly, set our **envelope** to linear (”Lin.”). A linear envelope will cause a more dramatic sound than a logarithmic (”Log.”) envelope, which adjusts the release time based on how sharp the transients are.

To bring back some stability, set **Lookahead Time** to 1ms and toggle on **Makeup Gain**. The 1ms lookahead gives a 1ms delay between receiving the signal and compressing it, while makeup gain automatically compensates the output. These two together keep our dynamics from being overly-squashed.

Finally, set the Output Gain to around -8dB to compensate for all the extra volume we’ve added with our previous parameters.

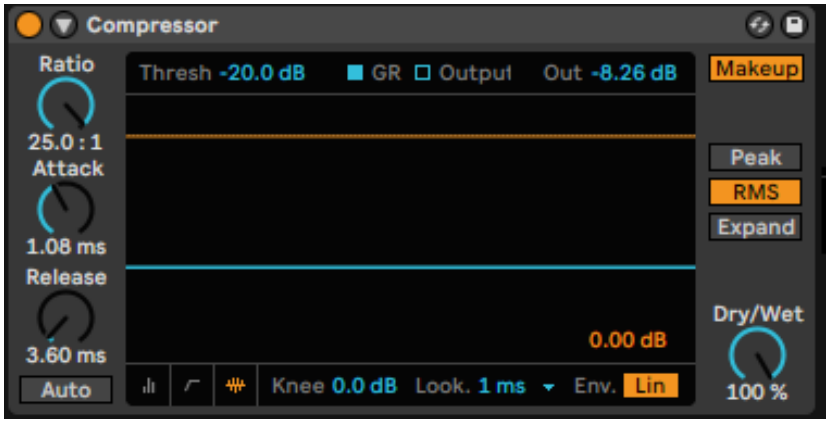

fig. 20 “The Suck” Compressor

With all of this intense compression, certain parts of the frequency spectrum can quickly get out of control and muddy up our mix. Luckily, we now have EQ Eight in our toolbox to deal with this exact problem. Add an EQ Eight after the compressor. Apply a low cut filter around 70Hz to cut out any low-frequency mud, some gentle cuts around 120Hz and 260Hz to allow any bass and kicks to keep more of their transients, and finally a high cut around 18Hz to tame any airy frequencies.

fig. 21 Taming mud with EQ

Now simply turn up the corresponding Send knob (in my case, C) on any MIDI or audio tracks that you want the effect on.

vii. Multiband Dynamics

Multi band Dynamics is simultaneously one of the most used and least understood tools in Ableton. Its effect can range from extremely subtle to completely transformative.

In essence multiband dynamics allows you to selectively split and compress different parts of the frequency spectrum. This allows you to increase or decrease the presence of lows, mids, and highs in any sound you put into it. In common usage, multiband dynamics is used for one thing only: OTT

The sound of Electronic Music: OTT

OTT (Over the Top) is one of the most iconic presets in Ableton and is one of the biggest contributors to the signature sound of much contemporary electronic music.

OTT, as the name implies, heavily reshapes any sound you put into it in often unpredictable ways. It is most commonly used on basses to give a heavily compressed in-your-face quality, and on high frequency sounds like hats to drastically increase their ability to cut through the mix. OTT can be used subtly with a low dry/wet mix, or stacked with 5+ OTTs at 100% wet in a row.

To access OTT, click on Multiband Dynamics in the File Browser to reveal the presets, then drag “OTT.adv” onto your track.

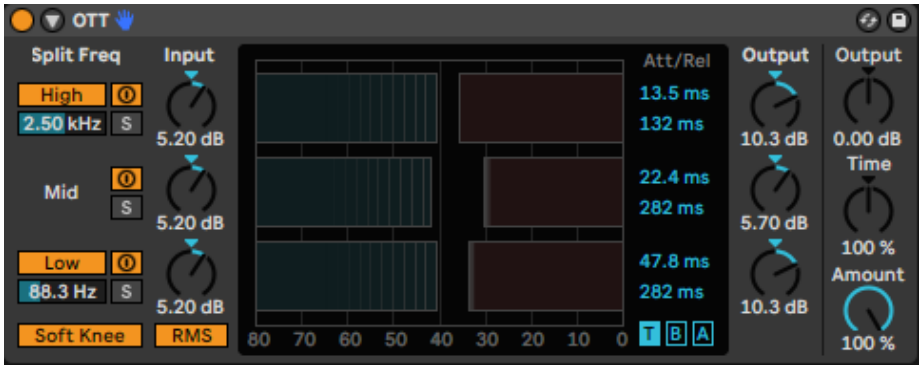

fig. 22 The Multiband Dynamics Preset “OTT”

Because we’re working with a preset, we’ll only take a look at a few essential parameters for controlling its effect. **“Amount”** functions as your Dry/Wet knob, and should be the last parameter to adjust. Turning our attention to the rest of the device, we can see that Multiband Dynamics contains 3 sections based on frequency (from the top): “High”, “Mid”, and “Low”, as is labeled along the left under **“Split Freq”**.

The high and low sections both contains **frequency sliders** underneath, which tell the device how to split up the frequencies. Adjusting these two frequency sliders is a good place to start to dial in OTT’s compression. For example, on a bass, you may want to adjust the high frequency to decide how much upper frequency content you want to stay, and vice versa with the low frequency for higher pitched sounds.

Finally, taking a look at the 3 knobs under the **“Output”** section, we can adjust the amount of compression happening based on the frequencies we just defined. Dialing these 3 knobs in using the same concept as the split frequencies, you can change how drastic OTT’s effect is, and subsequently how much noise is added in to your original sound.

viii. Grain Delay

Grain delay is a swiss army knife of sound design and one of the most fun effect devices to play around with. At its heart, grain delay is a granular resampler organized into a matrix. It will take any sound input, split it into grains determined by the parameters along the Y-axis, and allows you to quickly modify these parameters in a dynamic way by selecting any two parameters to form the X and Y-axis of the matrix.

Grain delay is great for:

- Adding texture to a sound
- Resampling with different frequency and spray values to layer and add complexity and movement to a sound
- Repitching a sound in unique ways
- Modulating with an LFO to produce ever-evolving textures

fig. 23 Grain Delay's user interface

Most of these parameters should be familiar from our exploration of granular synthesis last week. What is new here is a **"Rand Pitch"** parameter that randomizes the pitch of each grain, and a **"Feedback"** parameter to add feedback to the input.

The two parameters I most often change with Grain Delay are **"Spray"** and **"Frequency"**. If we select these two parameters on the matrix, setting our node toward the top right will produce a noisy, distorted sound almost like combining the original sound with white noise, while values toward the bottom left will produce wide, warble-y, chorus-like sounds.

Try adding LFOs to the effect chain with your Grain Delay and modulating Spray, Frequency, and Time (accessed by clicking the **"Sync"** button) while the Feedback parameter is above 50%. Using sharper waveforms for the LFO with a higher rate can produce great material for resampling and cutting up by letting the LFOs run and recording its contents into a new track.

ix. Auto Pan

Auto pan is an LFO that works on the stereo field only, and is a great effect for adding modulation to the **stereo field**.

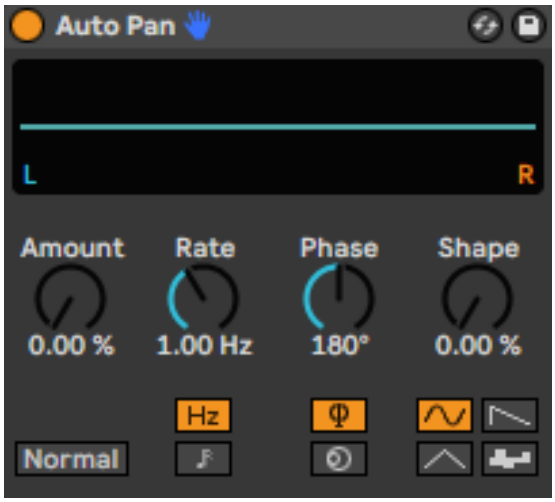

fig. 24 Auto Pan's user interface

The parameters of Auto Pan should be familiar from working with LFOs, but with two new parameters. **"Amount"** functions as a dry/wet knob, while **"Phase"** sets the phase between the left and right channels. You can see how this works by setting amount to 100%, then bringing phase down to 0°. Notice how the right channel (in yellow) overlaps more and more with the left channel (in blue).

One common way to use Auto Pan is to add very slow movement to pads or ambience to create subtle movement and space in a piece by using a small value for rate and amount. Using a rate in hertz that is slightly different between multiple layers will produce a large depth of movement and a sense of space to your pads or ambience.

Another common use is to add quantized movement to percussive elements like hats. Change the waveform to a saw wave to get clean movement with each cycle. Change the rate to notes, and set the rate to half the rate of the notes that are being played (e.g: for sixteenth note hats, set the rate to 1/8). This will make it so every other hat plays in the left and right, adding much more interest to your hats.

My favorite use of Auto Pan comes from using it to produce **risers**. Set "Amount" to 100%, "Phase" to 0.00°, "Shape" to 0.00%, and select the sawtooth waveform. Now bring "Rate" all the way up to 90.0 Hz, and smoothly drag it back down. This will

produce a **stutter effect** whose rate can be automated entirely with a single knob. For a more subtle effect, you can try setting amount to a lower value, or using a softer waveform like the triangle or sine waves.

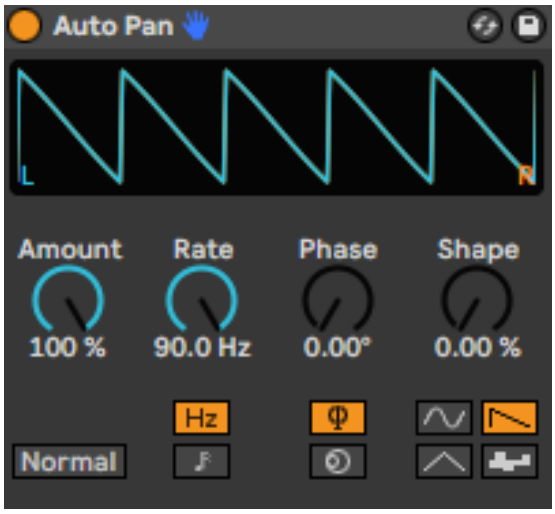

fig. 25 Auto Pan being used to produce a stuttering riser

2. MIDI Effects

i. Infinite Random Sample Selector with Drum Rack, Random, & Arpeggiator

To demonstrate two of the more useful MIDI effects in Ableton, let's build an infinite random sample generator using them.

To begin, place down a **drum rack**. Drum rack is an instrument that functions as a giant sample container that maps each sample to a MIDI note. You can automatically populate each note in your drum rack by dragging and dropping a bunch of samples that will automatically map to MIDI notes upon being dropped.

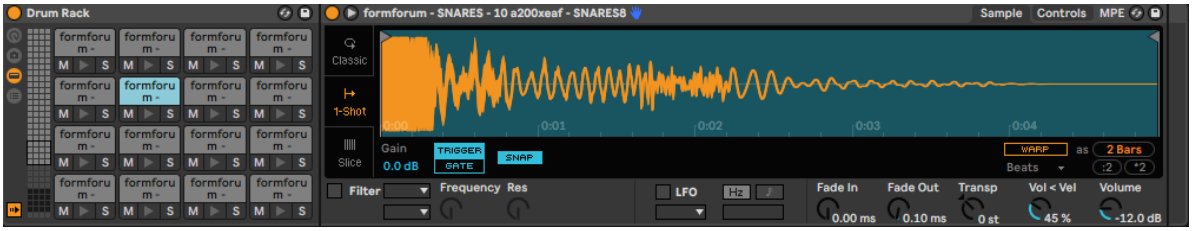

fig. 26 Drum rack populated with samples

You can play each sample by playing its corresponding note. To start adding randomness, we can use a **random** MIDI effect, which randomizes every MIDI note sent into it, so that we can play just a single note and have drum rack pick a random sample each time. Place down a Random device before Drum Rack from the “MIDI Effects” section of the file browser.

The only two parameters to worry about here are **“Chance”** and “**Choices”**. Chance is a measure of how often a different random note is played instead of the note you press. Set chance to 100% so we always get a new sample. Choices controls how many notes above the note you press can be randomly selected. Set this number to 24, or the number of samples you have if you have less than 24.

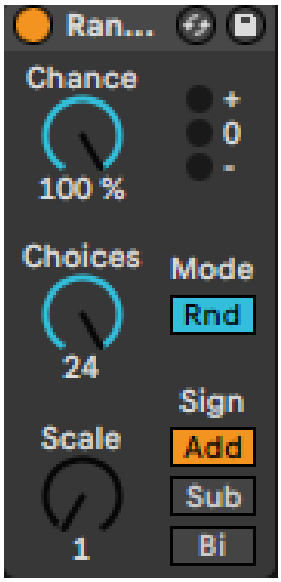

fig. 27 Random being used to trigger a random sample each time a note is pressed

Now to get it so we don’t have to play each note ourselves, add our second MIDI effect: **Arpeggiator** before Random. Arpeggiator is an incredibly powerful and flexible arpeggiator, letting you hold down notes while it plays each note in patterns to generate arpeggios. For our purposes, we only have to worry about the **“Rate”** and **“Steps”** parameters. Set the rate to whatever note length you want each sample to be triggered at, or click the **“Sync”** toggle to use a rate in milliseconds instead. Now if you have

more than 24 samples, increase steps, which will cause the arpeggiator to play a corresponding amount of octaves up. For a full drum rack, set steps to 6.

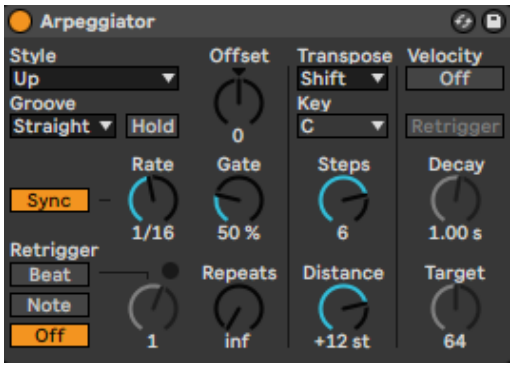

fig. 28 Arpeggiator being used to trigger a full drum rack of samples

Now as you hold down your root note (C1), you will hear an infinite stream of random samples being triggered at the rate you chose. This is a great technique to use on percussion to generate new rhythmic ideas. Try using extremely short samples (under 1sec long), perhaps resampled from a granular synthesizer, with a quick free rate to get an bed of glitchy IDM chaos. Resample chords down as audio files and use them as samples in your drum rack to generate chord progressions. Record out your stream of sampled audio and cut between samples in new ways, then feed them back in as new samples into the drum rack to generate infinite new samples as material. There's no limit to the amount of sounds you can get out of this technique!